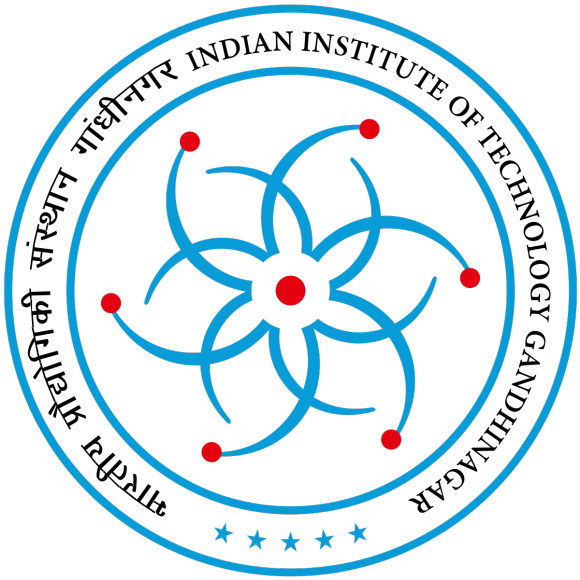
Indian Institute of Technology Gandhinagar



Four-Bar Mechanism Synthesis and Simulation for Upper-Arm Rehabilation

ME 335 Synthesis and Analysis of Mechanisms

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Date of Submission: 27/02/2026

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1 Introduction

Rehabilitation of the upper arm is essential for restoring mobility, strength, and functional independence in patients affected by stroke, fractures, neurological disorders, and muscular injuries. Conventional rehabilitation techniques primarily depend on manual therapy administered by physiotherapists. While effective, these methods are physically demanding, time-intensive, and often limited by the availability of trained professionals.

This project focuses on the design and simulation of a simple, reliable, and cost-effective mechanical system capable of guiding controlled upper arm motion. The proposed design aims to provide safe, efficient, and comfortable rehabilitation, making it suitable for both clinical and home-based applications.

2 Theoretical Synthesis

2.1 Objective

The objective is to design a mechanical system that can assist in upper arm rehabilitation by generating smooth, controlled, and repeatable motion, while ensuring patient comfort, operational safety, and mechanical feasibility.

2.2 Design Requirements

Since the mechanism is to be used for upper-arm rehabilitation, the mechanism type and its link lengths must be chosen so that they replicate the actual motion of an arm during certain exercises.

A four-bar mechanism was hence opted since it can provide various motions while still being a single DOF mechanism.

2.3 Mechanism type selection

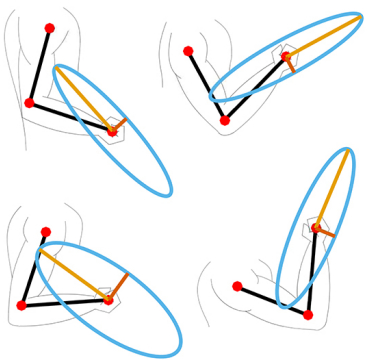


Figure 1: Human hand

Since we are designing a mechanism for upper-arm rehabilitation, we need to ensure that the wrist can rotate fully, whereas the shoulder only moves through a limited range. This helped us narrow down our and we selected a **crank-rocker mechanism** for our project.

2.4 Methodology for Data Collection

To obtain realistic human arm motion data for mechanism synthesis, an experimental motion capture study was conducted using a marker-based optical tracking system. Reflective markers were attached to key anatomical landmarks of the human subject, specifically at the shoulder, elbow, wrist, and finger locations. These markers enable precise three-dimensional position measurement over time.



Figure 2: Human subject with motion capture markers placed on the arm for data acquisition.

The subject was instructed to perform controlled arm movements representative of rehabilitation exercises. The motion capture system recorded the spatial trajectories of each marker in three-dimensional space at discrete time intervals, generating time-series positional data.

The collected data was exported in CSV format containing frame number, timestamp, and marker coordinates along the X, Y, and Z directions. These datasets were subsequently processed using Python for kinematic reconstruction and analysis.

2.5 Data Processing and Kinematic Reconstruction

The recorded marker data was processed to reconstruct the arm kinematics. Since two markers were available near the wrist, the wrist joint location was computed as the midpoint between the two wrist markers to improve positional robustness and reduce measurement noise.

The arm was modeled as a planar linkage consisting of four key points:

- Shoulder (SHO)
- Elbow (ELB)
- Wrist (WRI)
- Finger (FIN)

These points were connected sequentially to reconstruct the arm configuration at each time instant.

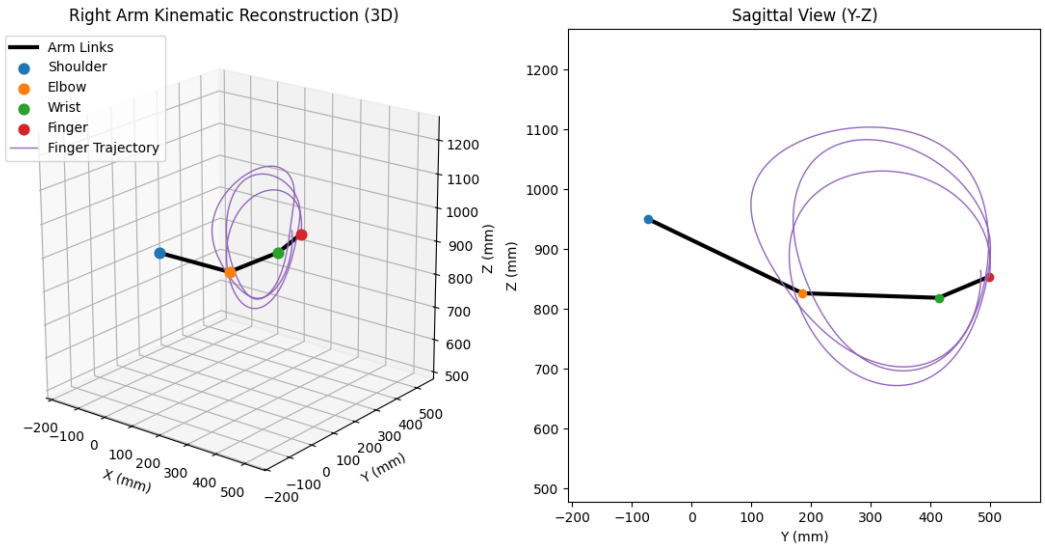


Figure 3: Three-dimensional reconstruction of the human arm motion and its sagittal plane projection.

For mechanism synthesis purposes, the motion was projected onto the sagittal plane (Y–Z plane), since rehabilitation arm motion primarily occurs in this plane. This simplification allows the human arm to be approximated as a planar four-bar mechanism.

2.6 Motion Analysis and Key Position Extraction

To identify representative positions for mechanism synthesis, several discrete frames were selected from the recorded motion sequence. These frames correspond to characteristic arm postures during the movement cycle.

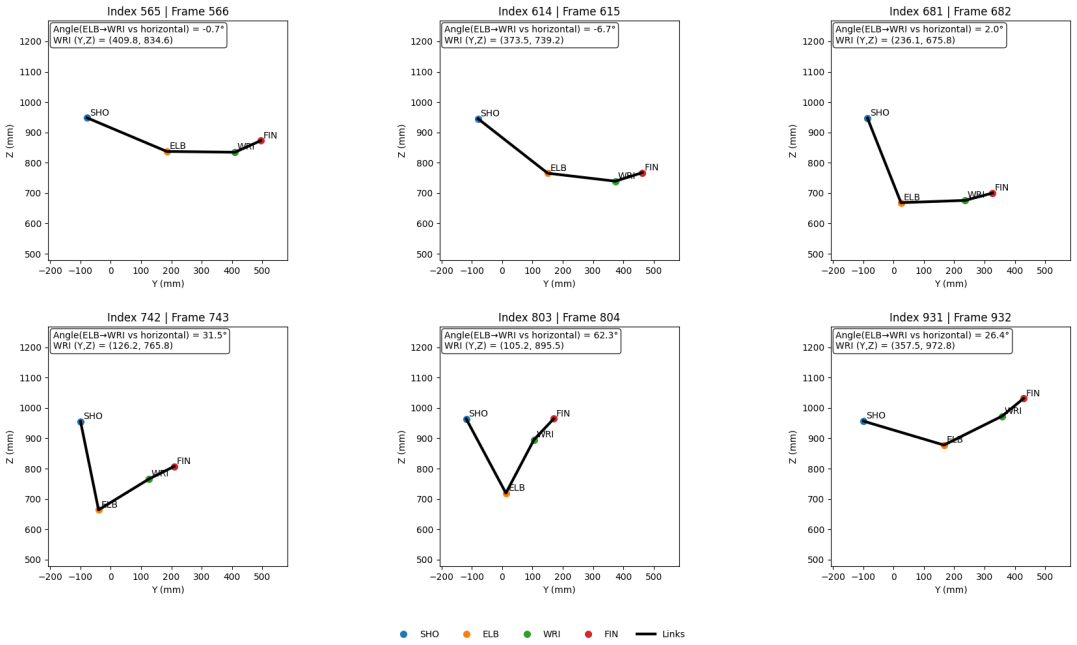


Figure 4: Selected sagittal plane arm configurations used for mechanism synthesis. Joint locations and link orientations are shown along with computed geometric parameters.

For each selected configuration, the following parameters were computed:

- Wrist coordinates in the sagittal plane
- Orientation angle of the forearm segment (elbow to wrist) with respect to the horizontal
- Relative link geometry between shoulder, elbow, wrist, and finger

The forearm orientation angle θ was computed using:

$$\theta = \tan^{-1}\left(\frac{Y_{\text{WRI}} - Y_{\text{ELB}}}{X_{\text{WRI}} - X_{\text{ELB}}}\right) \tag{1}$$

These extracted configurations serve as precision positions for the graphical and analytical synthesis of the rehabilitation mechanism.

The processed motion data therefore provides realistic geometric constraints that guide the determination of link lengths and motion characteristics of the proposed four-bar mechanism.

2.7 Graphical Synthesis

2.7.1 Coordinates and Graphical representation

From the plots shown above, we used the six configurations to obtain the coordinates of the wrist and elbow joints and the angle the forearm makes with the horizontal. The coordinates are shown in the table below.

Elbow x	Elbow y	Wrist x	Wrist y	Angle (°)
109.82	838.27	409.8	834.6	-0.7
75.55	774.2	373.5	739.2	-6.7
-63.72	665.33	236.1	675.8	2
-129.59	609.05	126.2	765.8	31.5
-34.25	629.88	105.2	895.5	62.3
88.79	839.41	357.5	972.8	26.4

Table 1: Coordinates of Elbow and Wrist

All dimensions are in mm and the length of fore-arm is taken to be equal to 30 cm.

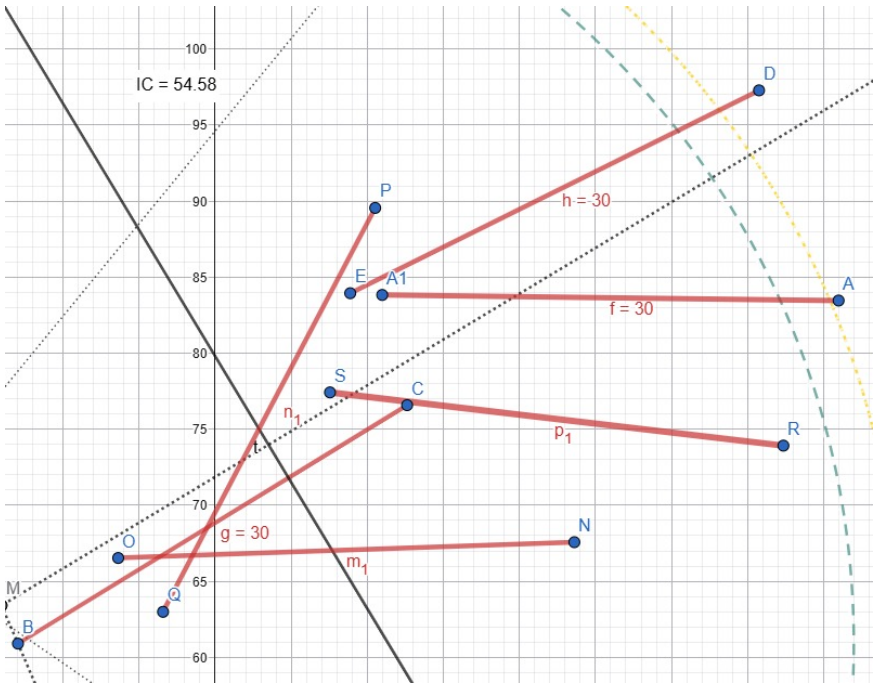


Figure 5: Coupler positions Sketched on GeoGebra

2.7.2 Advantage of Graphical Synthesis

While analytical synthesis offers high numerical precision, we chose Graphical Synthesis with fixed pivots because of the specific constraints of the upper-arm rehabilitation task. In analytical synthesis, especially with five-precision-point path generation, the loop-closure equations became very complicated. We did get link lengths, but the mechanism constructed from them was not satisfactory, so we went ahead with the graphical method. However, in subsequent submissions, we will implement the analytical method.

By choosing the Graphical Method, we gained several advantages:

- **Intuitive Constraint Management**: We could easily see how the mechanism was grounded, making sure the fixed pivots did not block the patient's seating area or movement.
- **Instant Error Detection**: Graphical construction allowed us to quickly spot Toggle Positions (singularities). If a linkage configuration got stuck between the

30° and 45° wrist positions, it was clear and could be fixed by adjusting the circle or center point curves.

- **Geometric Verification**: Since we used MotionGen, we were able to visualize that the coupler link's path remained smooth and natural, avoiding the jerky actions sometimes seen in purely numerical solutions that lack a smoothness constraint.

We used three coupler positions (ON, PQ, A1A) and fixed our ground pivots accordingly to ensure that the link lengths are physically possible and are not too big or small.

2.7.3 Procedure

We know three coupler positions C_1D_1, C_2D_2, C_3D_3 and the ground pivots are A_0, B_0.

Since the ground pivots A_0 and B_0 are fixed, we invert the motion by fixing the coupler in position C_1D_1 and determining the relative positions of the ground pivots with respect to it.

Step 1: Construct moving positions of ground pivots

- From positions C_2D_2 and C_3D_3, measure the distances from A_0 and B_0 to the coupler points.
- Keeping C_1D_1 fixed, reconstruct the corresponding locations of the ground pivots to obtain:
$$A_0',\ A_0'' \quad \text{and} \quad B_0',\ B_0''$$
- Thus, three positions of each ground pivot relative to the fixed coupler are obtained:
$$A_0,\ A_0',\ A_0'' \qquad B_0,\ B_0',\ B_0''.$$

Step 2: Locate moving pivot E

- Draw the perpendicular bisector of A_0A_0'.
- Draw the perpendicular bisector of A_0A_0''.

Their intersection gives point E, such that
$$A_0E_1 = A_0E_2 = A_0E_3.$$

Step 3: Locate moving pivot F

- Draw the perpendicular bisector of B_0B_0'.
- Draw the perpendicular bisector of B_0B_0''.

Their intersection gives point F, such that
$$B_0F_1 = B_0F_2 = B_0F_3.$$

Step 4: Form the four-bar mechanism
Since E and F lie on the same rigid body (the coupler), the four-bar mechanism
$$A_0 - E - F - B_0$$
is synthesized, where A_0E is the input link, EF is the coupler, FB_0 is the output link, and A_0B_0 is the fixed ground link.

2.7.4 Link Lengths Obtained

The link lengths obtained using graphical synthesis are shown below.

Ground link:- 22.5 *cm*
Crank link:- 12.87 *cm*
Rocker link:- 20.73 *cm*
Segment AB:- 16.32 *cm*
Segment BC:- 24.46 *cm*
Segment CD:- 13.81 *cm*
Segment AD:- 30.52 *cm*

2.8 Kinematic Diagram

The kinematic diagram for the obtained four bar mechanism is shown below.

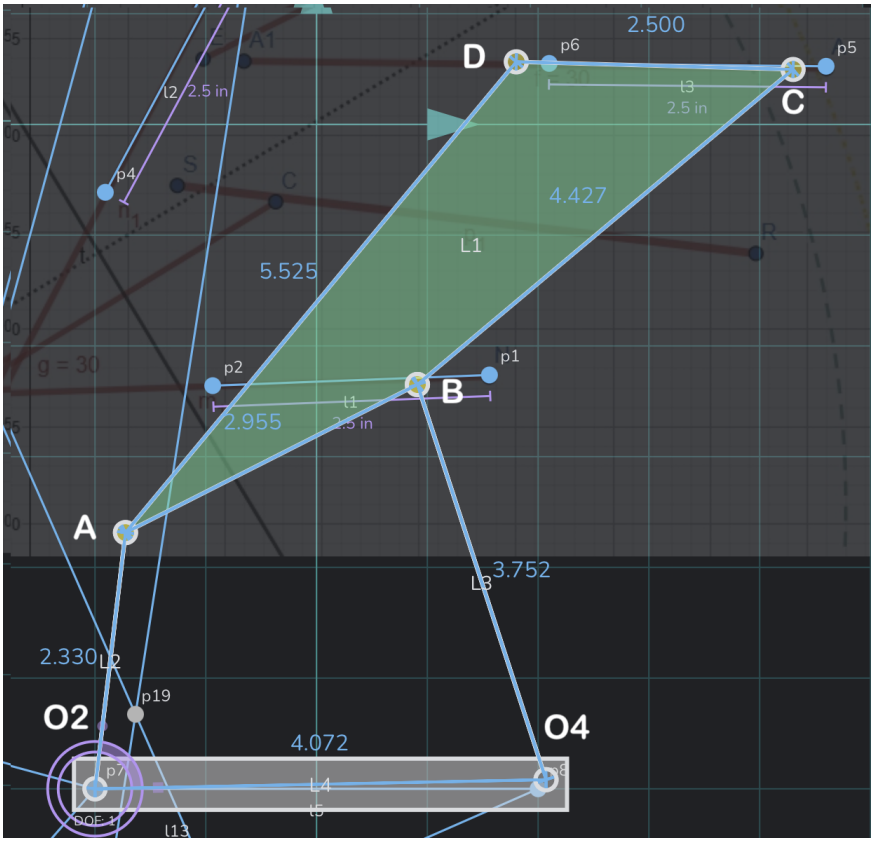


Figure 6: Kinematic Diagram

2.9 Software-based simulation

The simulation of the mechanism was done on MotionGen and the video of the simulation is present in the google drive link submitted.

2.10 Feasibility Check

2.10.1 Simulation Results

During the simulation, the mechanism moved smoothly without link interference. All joints remained properly constrained, and the output link followed the required oscillatory path for the patient's arm movement. The obtained motion matched the intended design.

2.10.2 Mobility and Grashof Condition

The mechanism satisfies the Grashof condition:

$$s + l < p + q \tag{2}$$

In our case the value of $s + l = 29.19\ cm$ and the value of $p + q = 43.23\ cm$

where s is the shortest link, l is the longest link, and p and q are the remaining links and hence it satisfies the above condition.

Since the shortest link is adjacent to the ground link, the mechanism works as a crank–rocker. The input link completes a full 360° rotation, while the output link oscillates between two fixed limits.

This configuration is suitable for rehabilitation because a continuously rotating motor can be used as input, and the rocker motion naturally limits the range of movement for safety.

2.10.3 Mechanical Feasibility

The link lengths obtained from graphical synthesis are practical for manufacturing. Standard pin joints can be used, and no interference occurs during motion. The mechanism can be fabricated using common materials such as aluminum or mild steel.

Thus, the design is both kinematically correct and mechanically feasible for physical realization but further improvements can be made to the mechanism.

3 Conclusion and Future Work

A mechanical system for upper arm rehabilitation was successfully designed through theoretical synthesis and kinematic simulation. The four-bar linkage mechanism demonstrated smooth, stable, and ergonomically suitable motion for therapeutic applications but more optimization can still be done as we observed that during the motion there are chances of getting a jerk which is not acceptable when creating a mechanism for rehabilitation. Future enhancements will include an improved version of the mechanism and dynamic force analysis, integration of motorized actuation, implementation of sensor-based feedback control, and experimental validation through clinical trials to further improve system performance.

4 Acknowledgments

We would like to express our gratitude to Professor Vineet Vashista for teaching us the concepts of Synthesis of Mechanisms clearly and concisely, and for assigning this project

to practically test our understanding of the concepts we were taught. Along with this, I would like to thank our Teaching Assistants, Mr Vardan Mittal and Mr Aaditya Nimkar who helped us for collecting the required data using motion capture and clarified any concerns we had while completing this project.

5 References

- Image-1 link
- Github Link for Project
- Google Colab Link (To access Python Code)
- Google drive link (To access entire folder)